



PromptPaint: Implementations and Contributions

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The Challenge

- Text prompts are imprecise, concepts that are hard to verbalize.
 - a style that's "between impressionism and picasso" difficult to explain with natural language
- T2I models are end-to-end.
 - prompt goes in, image comes out, no way to intervene or steer during the process
- No spatial control.
 - the same prompt applies to the entire canvas
- Iteration is blind.
 - There's no way to know why or where to adjust; you just re-prompt and hope

The Theory Behind PromptPaint

- Prompts are just vectors in an embedding space.
 - Vectors can be mixed, shifted, and blended.
 - Navigate the semantic space directly by manipulating those vectors
- PromptPaint maps these properties onto paint-medium interactions:
 - mixing → blending prompts
 - brushing → spatial control
 - Multiple strokes → prompt intervention

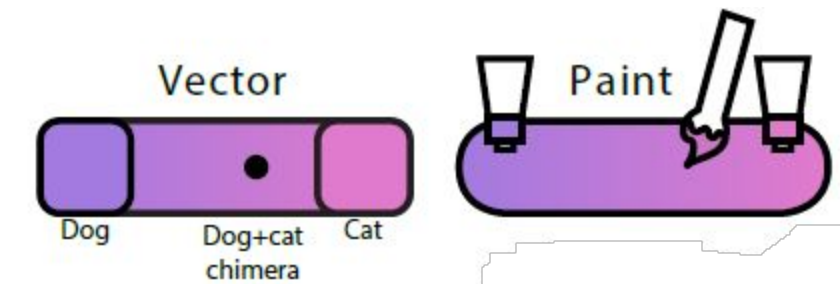


Fig 1.Paint Mixing

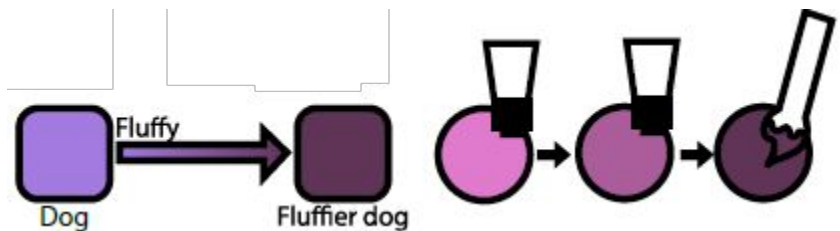


Fig 2.Paint Shifting

My Main Contributions

- Fréchet Mean Prompt Mixing instead of Linear Interpolation
- Upgraded to SDXL instead of SD v1.5
- Fully post-hoc generation controller instead of live prompt intervention
- Prompt Mixing works for more than 3 prompts
- Fully deployed webapp product

Fréchet Mean Prompt Mixing

Problem:

- CLIP embeddings live on a high-dimensional hypersphere.
- Naive linear interpolation shrinks the resulting vector, pulling it off the sphere to out-of-distribution region.

Fréchet Mean Prompt Mixing

- The Fréchet mean finds point on sphere that minimizes the sum of squared distances to all input prompts
- Result: a mixed embedding that stays on the hypersphere, is order-independent, and correctly handles N-way mixing

PAPER	MY METHOD
weighted sum $v_{pm} = \sum_{i=1}^N w_{p_i} v_{p_i}$	Fréchet mean objective $v_{pm} = \arg \min_{v \in S} \sum_i w_{p_i} d_g(v, v_{p_i})^2$ iterative update (log / exp maps) $v_{pm} \leftarrow \exp_{v_{pm}} \left(\sum_i w_{p_i} \log_{v_{pm}}(v_{p_i}) \right)$

Fig 3. Differences between two approaches

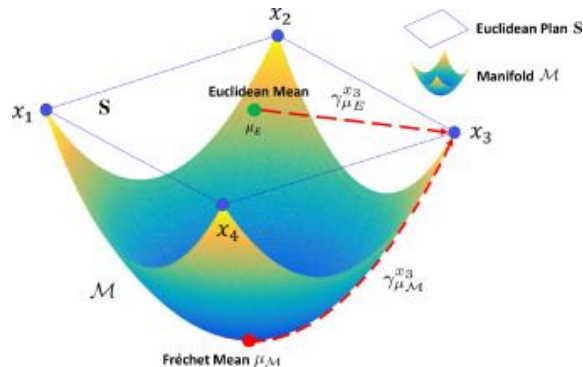


Fig 4. Frechet Mean Prompt Mixing

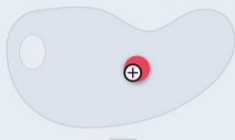
PROMPT LIST

+ Add

1 Prompt 1...

PROMPT PALETTE ①

Fixed Edit



DIRECTIONAL PROMPTS ①

+ Add

From... To...
→ To 1.0×

PROMPT COMPOSITION

1 100%

• Painting to full canvas

Paint



LAYERS



+ @

Sizing 16:9 L

+ 82% - Reset

PROMPT LIST

+ Add

1 Prompt 1...

PROMPT PALETTE ①

Fixed Edit



DIRECTIONAL PROMPTS ①

+ Add

From... To...
— To 1.0x

PROMPT COMPOSITION

1 100%

• Painting to full canvas

Paint



Welcome to PromptPaint

Your Canvas

Generated image will appear here

LAYERS



+ ⌵

Sizing 16:9 L

+ 82% - Reset



PROMPT LIST

+ Add

1 Prompt 1...

PROMPT PALETTE ①

Fixed Edit



DIRECTIONAL PROMPTS ①

+ Add

From... To...

→ To 1.0x

PROMPT COMPOSITION

1 100%

• Painting to full canvas



Paint



Sizing 16:9 L

+ 82% - Reset

PROMPT LIST

+ Add

1 Prompt 1...

PROMPT PALETTE ①

Fixed Edit



DIRECTIONAL PROMPTS ①

+ Add

From... To...

→ To 1.0x

PROMPT COMPOSITION

1 100%

• Painting to full canvas

Paint



Welcome to PromptPaint

Your Canvas

Generated image will appear here

LAYERS



+ ⌵

Sizing 16:9 L

+ 82% - Reset

PROMPT LIST

+ Add

1

Prompt 1...

PROMPT PALETTE ①

Fixed Edit



DIRECTIONAL PROMPTS ①

+ Add

From...

To...

→ To 1.0x

PROMPT COMPOSITION

1

100%

● Painting to full canvas

Paint

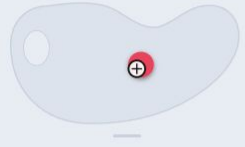
PROMPT LIST

+ Add

1 Prompt 1...

PROMPT PALETTE ①

Fixed Edit



DIRECTIONAL PROMPTS ①

+ Add

From... To...
→ To 1.0x

PROMPT COMPOSITION

1 100%

GENERATION SETTINGS

STEPS — 40 GUIDE SCALE — 7

SINGLE STROKE — 20%

STENCIL Off

BRUSH RADIUS — 50px OVERCOAT — 100%

• Painting to full canvas

Paint



LAYERS



+ -

Sizing 16:9 L

+ 80% - Reset



PROMPT LIST

+ Add

1 Prompt 1...

PROMPT PALETTE ①

Fixed Edit

DIRECTIONAL PROMPTS ①

+ Add

From... To...

→ To 1.0x

PROMPT COMPOSITION

1 100%

GENERATION SETTINGS

STEPS — 40 GUIDE SCALE — 7

SINGLE STROKE — 20%

STENCIL Off

BRUSH RADIUS — 50px OVERCOAT — 100%

● Painting to full canvas

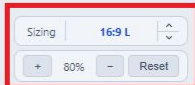


Welcome to PromptPaint

Your Canvas
Generated image will appear here

Sizing 16:9 L

+ 80% - Reset



PROMPT LIST + Add

1 Prompt 1...

PROMPT PALETTE ① Fixed Edit

DIRECTIONAL PROMPTS ① + Add

From... To...

→ To 1.0x

PROMPT COMPOSITION

1 100%

GENERATION SETTINGS

STEPS — 40 GUIDE SCALE — 7

SINGLE STROKE — 20%

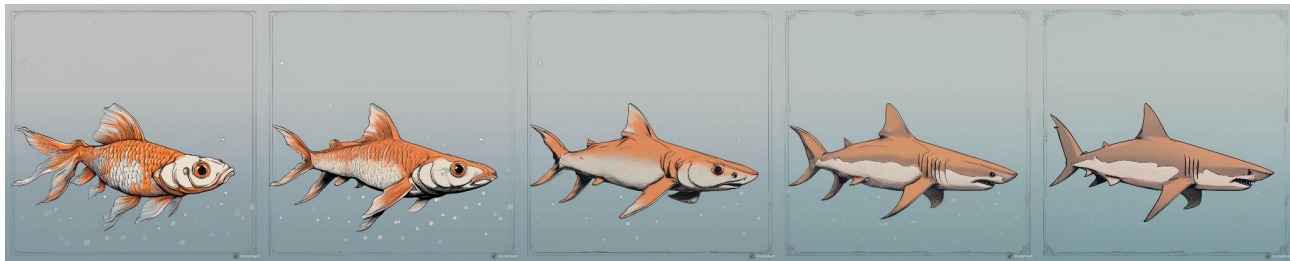
STENCIL Off

BRUSH RADIUS — 50px OVERCOAT — 100%

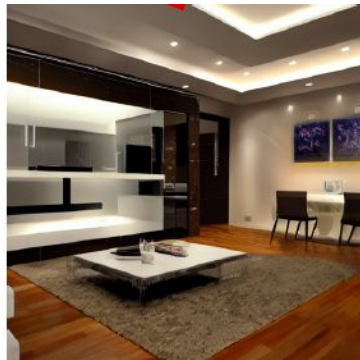
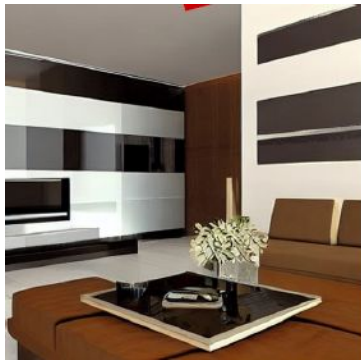
• Painting to full canvas

Paint

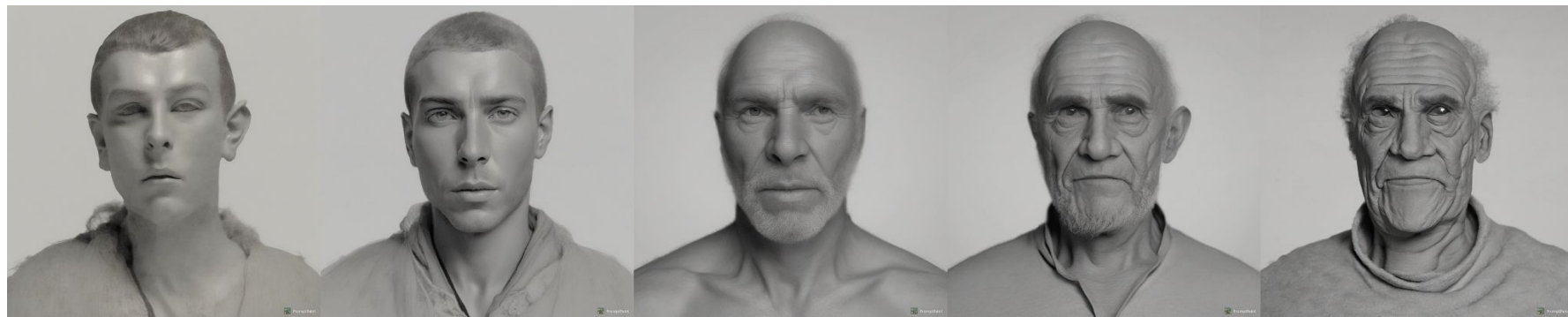
Prompt Mixing



Directional Prompt: Interior home design



Directional Prompts



Prompt Intervention: rabbit -> ceramic cup

No Switching

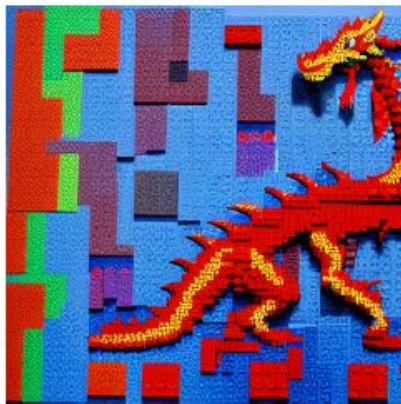
Switching at 75%

Switching at 50%

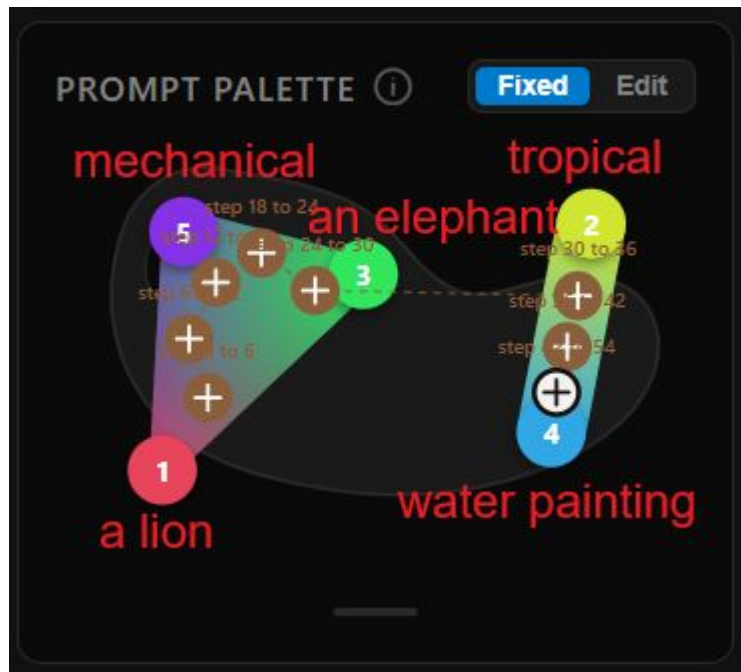
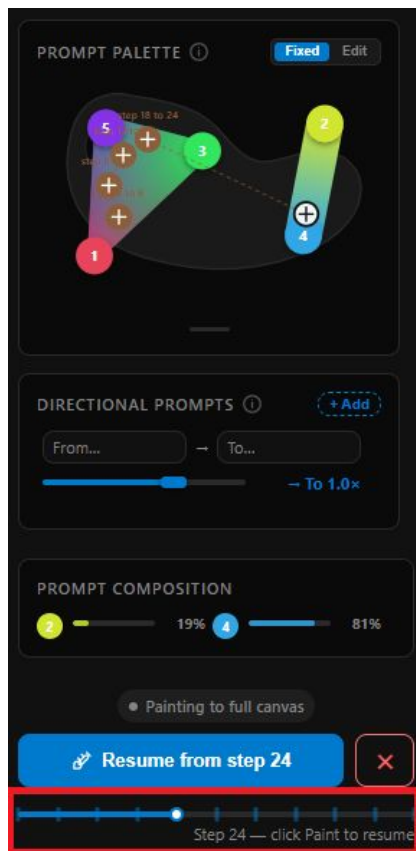
Switching at 25%



Prompt Intervention: Action painting -> lego dragon



Prompt Intervention Mechanism





Why the SDXL 1.0

- This webapp hosted by a dual GPU workstation
- SDXL requires 8gb of vram (out of 32gb)
- Lightweight model → always-on hosting
- Webapp currently at promptpaint.tom-nguyen.ca

